

Unix Workflow

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1. Where am I?

Normal session at TACC

```
1 login1.stamped3(107)$ ls
2  [ some output ]
3 login1.stamped3(108)$ mkdir somewhere
4 login1.stamped3(109)$ cd somewhere
5 login1.stamped3(110)$ ls
6  [ some output ]
```

- Problem: no idea what directory you are in.
- Solution: redefine command prompt

```
1 login1.stamped3(117)$ export PS1="[ \h \W:\#] "
2 [login1 ~:2] cd somewhere/
```

- Edit your `.bashrc` and add this line to the bottom
(and `source ~/.bashrc` the first time)

2. What are my files?

```
1 [login1 somewhere:5] touch onefile
2 [login1 somewhere:6] mkdir onedir
3 [login1 somewhere:7] ls
4 onedir onefile
5 onedir onefile
6 [login1 somewhere:11] alias ls='ls -F'
7 [login1 somewhere:12] ls
8 onedir/ onefile
```

- Use `alias` to supply standard options to common commands
- Again, put that in your `.bashrc`
- Example: case-insensitive search:

```
1 alias grep='grep -i'
```

3. Jumping between directories

```
1 [login1 ~:18] cd class_repository/  
2 [login1 class_repository:19] ls  
3 examplefile.cpp  
4 [login1 class_repository:20] pushd ../private_repository/  
5 ~/private_repository ~/class_repository  
6 [login1 private_repository:21] ls  
7 [login1 private_repository:22] pushd  
8 ~/class_repository ~/private_repository  
9 [login1 class_repository:23] ls  
10 examplefile.cpp  
11 [login1 class_repository:24] cp examplefile.cpp ../private_repository/  
12 [login1 class_repository:25] pushd  
13 ~/private_repository ~/class_repository  
14 [login1 private_repository:26] mv examplefile.cpp homework1.cpp
```

- `cd` to one directory,
- `pushd` to other directory
- `pushd` without arguments to switch
- `popd` if you're done with a directory

4. More jumping

Remember directory locations:

```
1 [login1 ~:33] export CLASSREPO=~/.class_repository
2 [login1 ~:34] export MYHWREPO=~/.private_repository
3 [login1 ~:42] cd $CLASSREPO
4 [login1 class_repository:43] pushd $MYHWREPO
5 ~/.private_repository ~/.class_repository
```

- Give names to paths with `export`
- and put those definitions in your `.bashrc`